2. The Chip Level

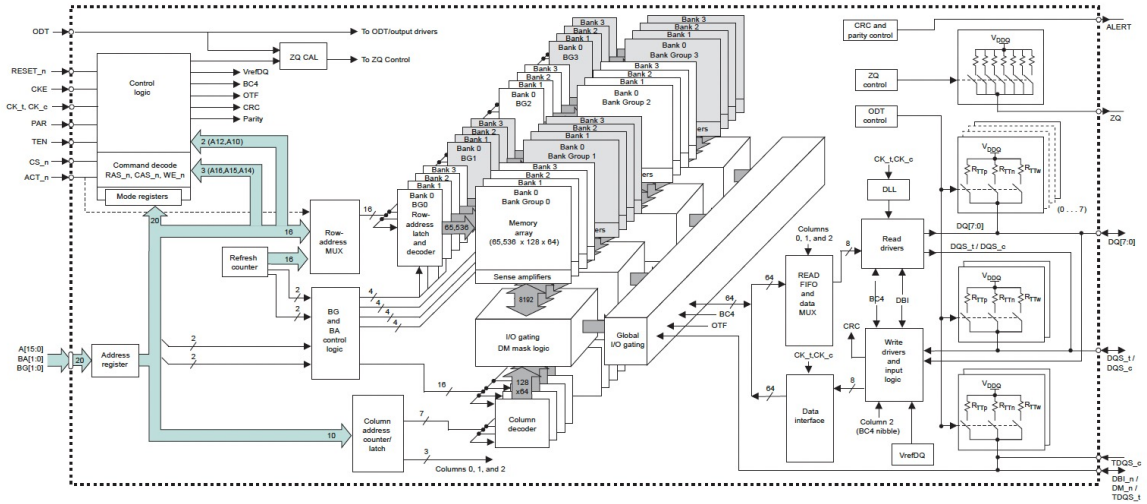
SDRAM

Sources:
- Balasubramonian, Rajeev. Innovations in the Memory System, Springer, 2019.

Micron DDR4 SDRAM MT40A1G8

- DRAM configuration is 1 Gig × 8.
  - 1 Gi ($= 2^{30}$) supercells, each providing 8 bits of data.
- Internally configured as a 16-bank, 4-bank group DRAM.
  - # bank groups = 4, BG[1:0].
  - # banks/group = 4, BA[1:0].
  - Row addresses = 64 Ki, A[15:0].
  - Column addresses = 1Ki, A[9:0]
  - Each bank has its own sense amps.
- Programming Mode Registers allow initialization and configuration of various parameters.

Figure 3: 1 Gig x 8 Functional Block Diagram



SDRAM Controller

- From manual:
  "This simplified state diagram provides an overview of the possible state transitions and the commands to control them. Situations involving more than one bank, the enabling or disabling of on-die termination, and some other events are not captured in full detail."
- After initialization, stays in Idle state.
- Automatically moves to Refresh state if needed.
- When a read/write request is made, moves via Bank active state to Reading or Writing state.
- After completing the request, controller moves to Idle, Bank active, or Precharging state.

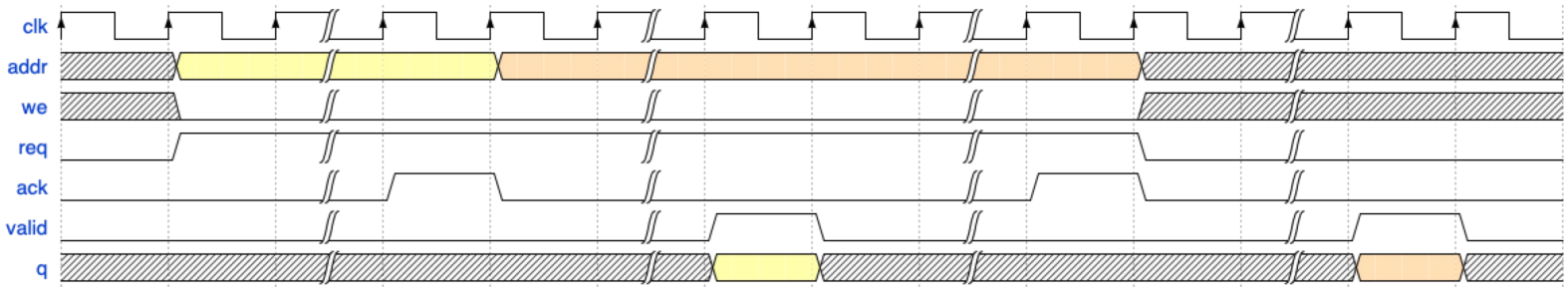
Figure 14: Simplified State Diagram

SDRAM Signals

Name	Direction	Description
`reset`	Input	Resets SDRAM controller when asserted.
`clk`	Input	System clock for SDRAM controller.
`addr`	Input	Address bus.
`data`	Input	Input data bus.
`req`	Input	Requests a read or write operation to be performed.
`we`	Input	Write enable: Asserted along with req to write to SDRAM.
`ack`	Output	Asserted when a request is accepted by SDRAM controller.
`valid`	Output	Asserted when there is valid data on output data bus.
`q`	Output	Output data bus.

SDRAM Controller

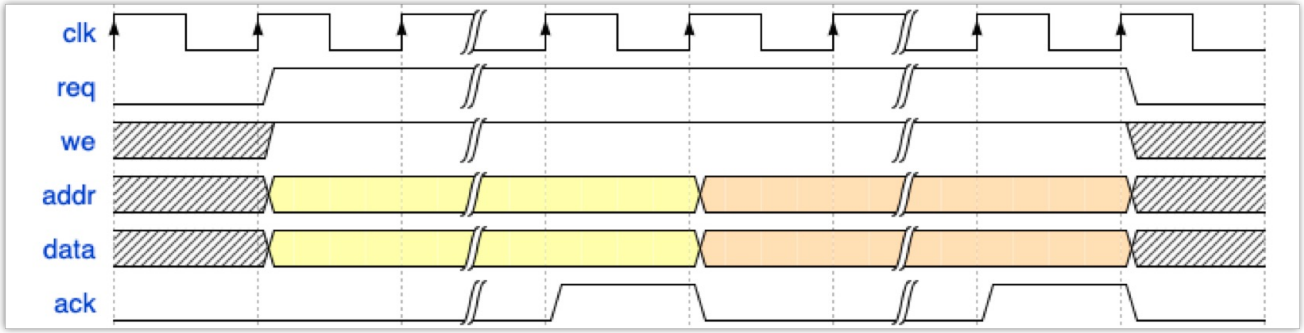
READ Timing Including Chaining



- Write address to `addr` bus. Request a read by deasserting `we` and asserting `req`. Wait for `ack` to be asserted. (READ #1)
- Write another address to `addr` bus. Request a read by deasserting `we` and asserting `req`. Wait for `ack` to be asserted. (READ #2)
- Wait for `valid` to be asserted. Read value on `q` data bus. (READ #1)
- Wait for `valid` to be asserted. Read value on `q` data bus. (READ #2)

SDRAM Controller

WRITE Timing Including Chaining



- Write address to `addr` bus and a value to the `data` bus. Request a write by asserting `we` and `req`.
- Wait for `ack` to be asserted. (WRITE #1)
- Write another address to `addr` bus and a value to the `data` bus. Request a write by asserting `we` and `req`.
- Wait for `ack` to be asserted. (WRITE #2)

Double Data Rate (DDR) Architecture

- DDR4 is an $8n$-prefetch architecture with an interface designed to transfer two data words per clock cycle at the I/O pins.
  - A single read or write access for a device module effectively consists of a single $8n$-bit-wide, four-clock-cycle-data transfer at the internal DRAM core and eight corresponding n-bit-wide, one-half-clock-cycle data transfers at the I/O pins.
- Basic idea
  - Assume CPU wants successive columns of a row. (Why?)
  - Read a fixed number of such columns from the row *together*; place them in a pre-fetch buffer.
  - I/O works on a faster clock and transfers at both `posedge` and `negedge`.

- $1024\,\frac{\text{col}}{\text{row}} \times 8\,\frac{\text{b}}{\text{col}} = \frac{1024}{8}\,\frac{\text{col grp}}{\text{row}} \times 8\,\frac{\text{col}}{\text{col grp}} \times 8\,\frac{\text{b}}{\text{col}} = 128 \times 64.$

  Burst Length
- Transfer rate
  - Part is MT40A1G8-062E.
  - Corresponding data rate is 3200 MT/s [from data sheet]
  - This corresponds to an I/O pin clock rate of 1600 MHz.
  - This corresponds to a DRAM core clock rate of 400 MHz.
- Peak transfer rate *for a DIMM built out of such parts* =

$$3200\times10^6\,\frac{\text{b}}{\text{s}} \times 64 \times \frac{1\text{ B}}{8\text{ b}} = 25.6\frac{\text{GB}}{\text{s}}.$$

Burst-Oriented Access

- Data is returned to the processor starting at the specified supercell. (Why?)
- Order of remaining supercells in FIFO is transmitted in order specified by burst type (BT) and burst length (BL).
  - Specified in Mode Register.
- Burst Type
  - Sequential.
  - Interleaved.
- Only relevant for read operations.

Table 8: Burst Type and Burst Order

Note 1 applies to the entire table

Burst Length	READ/ WRITE	Starting Column Address (A[2, 1, 0])	Burst Type = Sequential (Decimal)	Burst Type = Interleaved (Decimal)	Notes
BC4	READ	0 0 0	0, 1, 2, 3, T, T, T, T	0, 1, 2, 3, T, T, T, T	2, 3
		0 0 1	1, 2, 3, 0, T, T, T, T	1, 0, 3, 2, T, T, T, T	2, 3
		0 1 0	2, 3, 0, 1, T, T, T, T	2, 3, 0, 1, T, T, T, T	2, 3
		0 1 1	3, 0, 1, 2, T, T, T, T	3, 2, 1, 0, T, T, T, T	2, 3
		1 0 0	4, 5, 6, 7, T, T, T, T	4, 5, 6, 7, T, T, T, T	2, 3
		1 0 1	5, 6, 7, 4, T, T, T, T	5, 4, 7, 6, T, T, T, T	2, 3
		1 1 0	6, 7, 4, 5, T, T, T, T	6, 7, 4, 5, T, T, T, T	2, 3
		1 1 1	7, 4, 5, 6, T, T, T, T	7, 6, 5, 4, T, T, T, T	2, 3
	WRITE	0, V, V	0, 1, 2, 3, X, X, X, X	0, 1, 2, 3, X, X, X, X	2, 3
		1, V, V	4, 5, 6, 7, X, X, X, X	4, 5, 6, 7, X, X, X, X	2, 3
BL8	READ	0 0 0	0, 1, 2, 3, 4, 5, 6, 7	0, 1, 2, 3, 4, 5, 6, 7	
		0 0 1	1, 2, 3, 0, 5, 6, 7, 4	1, 0, 3, 2, 5, 4, 7, 6	
		0 1 0	2, 3, 0, 1, 6, 7, 4, 5	2, 3, 0, 1, 6, 7, 4, 5	
		0 1 1	3, 0, 1, 2, 7, 4, 5, 6	3, 2, 1, 0, 7, 6, 5, 4	
		1 0 0	4, 5, 6, 7, 0, 1, 2, 3	4, 5, 6, 7, 0, 1, 2, 3	
		1 0 1	5, 6, 7, 4, 1, 2, 3, 0	5, 4, 7, 6, 1, 0, 3, 2	
		1 1 0	6, 7, 4, 5, 2, 3, 0, 1	6, 7, 4, 5, 2, 3, 0, 1	
		1 1 1	7, 4, 5, 6, 3, 0, 1, 2	7, 6, 5, 4, 3, 2, 1, 0	
	WRITE	V, V, V	0, 1, 2, 3, 4, 5, 6, 7	0, 1, 2, 3, 4, 5, 6, 7	3

Notes:
1. 0...7 bit number is the value of CA[2:0] that causes this bit to be the first read during a burst.
2. When setting burst length to BC4 (fixed) in MR0, the internal WRITE operation starts two clock cycles earlier than for the BL8 mode, meaning the starting point for ^{t}WR and ^{t}WTR will be pulled in by two clocks. When setting burst length to OTF in MR0, the internal WRITE operation starts at the same time as a BL8 (even if BC4 was selected during column time using A12/BC4_n) meaning that if the OTF MR0 setting is used, the starting point for ^{t}WR and ^{t}WTR will not be pulled in by two clocks as described in the BC4 (fixed) case.
3. T = Output driver for data and strobes are in High-Z.
   V = Valid logic level (0 or 1), but respective buffer input ignores level on input pins.
   X = "Don't Care."